

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 65

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2's S0/0/1 interface is still configured in passive mode (only the S0/0/0 interface is enabled, passive mode is not enabled). Therefore, R2 will not send OSPF Hello packets through the S0/0/1 interface, and thus cannot establish an adjacency with R3. R3 must route to the 192.168.2.0/24 network segment through R1, not directly through R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

enable

configure terminal

router ospf 1

no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

Calculated as: Serial link cost (64) + R2's G0/0 cost (1) = 65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth in OSPF is 100 Mbps. With this setting, all connections with speeds of 100 Mbps or higher have a cost of 1, so Fast Ethernet, Gigabit Ethernet, and 10 Gigabit Ethernet connections appear the same in OSPF.

Step 2: Change the bandwidth for an interface.

- g. Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost of 192.168.3.0/24:

R1 S0/0/1 + R3 G0/0 (781+1 = 782).

The cost of 192.168.23.0/30:

R1S0/0/1and R3 S0/0/1 (781+ 64 = 845).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology. What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562.

After executing the 128-bandwidth instruction on all remaining sequence interfaces, the cost per sequence interface becomes 781 (100,000 ÷ 128).

Accessing the 192.168.23.0/30 network from R1 requires two sequence links:

Path via R2: R1's S0/0/0 (781) + R2's S0/0/1 (781) = 1562

Path via R3: R1's S0/0/1 (781) + R3's S0/0/0 (781) = 1562

Step 3: Change the route cost.

- c. Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF selects the path with the lowest cost. The cost of the path through R2 is: R1-S0/0/0 (781) + R2-S0/0/1 (781) + R3-G0/0 (1) = 1563. The cost of the path directly through R3 is: R1-S0/0/1 (1565, manually configured) + R3-G0/0 (1) = 1566. Since 1563 < 1566, OSPF chooses the lower cost path through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling router ID allocation prevents unexpected OSPF process restarts when physical interface changes occur, thus ensuring network stability. Using loopback interfaces or the `router-id` directive ensures router ID consistency and predictability, which not only simplifies troubleshooting but also prevents adjacency failures due to ID duplication and provides better control over DR/BDR election.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election process is unnecessary in this experiment because all OSPF neighbor relationships are established through point-to-point sequence links, and point-to-point serial links do not require DR/BDR election.

3. Why would you want to set an OSPF interface to passive?

Setting the LAN interface to passive mode prevents routers from transmitting OSPF Hello packets and route updates through this interface. This reduces unnecessary network traffic, saves bandwidth, improves security by not exposing routing information to end hosts, and prevents routers from wasting resources trying to establish adjacencies with non-existent OSPF neighbors. Connected networks are still advertised to other OSPF routers and are therefore still reachable.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The aspect of OSPF that makes the protocol hierarchical and permits the creation of very scalable networks is its area-based design.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Router(config-if) #ip address 10.0.0.0 255.255.255.0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

It will use the EIGRP route.

Preset administrative distances:

EIGRP: 90, OSPF: 110, RIP: 120

Because the administrative distance of EIGRP (90) is less than that of OSPF (110) and RIP (120), the router will choose the EIGRP route, even though its metric (4039043) is much greater than the other routes. Metrics are only compared when routes come from the same routing protocol.